Lamia Salsabil

Ph.D. Candidate | AI Engineering with LLMs, Multimodal LLMs, NLP | Scholarly Big Data

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github.com/LiyaLamia | linkedin.com/in/lamia-salsabil-94719b154 | Google Scholar

Education

Old Dominion University — *Norfolk, VA*

Ph.D., Computer Science (GPA: 3.77) — Dec. 2027 (Expected)

Advisor: Dr. Jian Wu, Candidacy passed Spring 2024

Khulna University of Engineering & Technology (KUET) — *Khulna, Bangladesh*

B.Sc., Computer Science and Engineering — Feb. 2018

Technical Skills

Generative AI: Proprietary LLMs (GPT, Gemini, Claude), open-weight LLMs (Qwen-VL, InternVL, MiniCPM-V, Mistral), multimodal LLMs, prompt engineering

ML Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas, spaCy

Document AI & OCR: GROBID, PyMuPDF, olmOCR, Amazon Textract, Tesseract

Data & Compute: PostgreSQL, MySQL, MongoDB, Apache Spark, Hadoop, AWS (S3), GPU computing, HPC (SLURM)

Languages: Python, Java, C++, C, R, SQL, JavaScript, Bash

Research Experience

Old Dominion University — *Norfolk, VA*

Graduate Research Assistant — *Jan. 2021 – Present*

Contextualized URL Extraction | *Multimodal LLMs, GPT-5-mini, Qwen3-VL* [Code]

- Created **sURL-3K**, the first benchmark for evaluating URL and context extraction, with 3,376 manually annotated URLs and their contexts from 352 scholarly papers.
- Built **VisURLX**, a multimodal LLM pipeline that reads PDF page images and extracts each URL with its context across the body, footnotes, and references, restoring footnote and reference entries into the sentences that cite them.
- Achieved **86.86% end-to-end F1**, 19–40 points above GROBID, olmOCR, and PyMuPDF pipelines.
- Presented the **first systematic comparison of image-based and text-based extraction** across accuracy, monetary cost, and computational cost, using proprietary and open-weight backends.

Dataset and Software URL Classification | *SciBERT, BertGCN, contrastive learning* [Code]

- Created **OADS-1K**, a benchmark for evaluating dataset and software URL classification, with 1,129 manually annotated URLs and their contexts from 1,574 scholarly documents.
- Developed **EnSU**, an ensemble of fine-tuned language models that assigns each URL one of six fine-grained classes from its context, crossing resource type (dataset or software) with provider (author or third party).
- Achieved **90% macro F1**, 15–22 points above few-shot GPT-4 and Claude baselines.

Computational Reproducibility Study | *DistilBERT, spaCy* [Code]

- Built a binary classifier that detects dataset and software URLs (**92% F1**) and used it on 100K ETDs to study disciplinary dependencies and chronological trends in including open datasets and software, later applied to 1.58M arXiv preprints.

Large-Scale Dataset and Software URL Repository | *VisURLX, EnSU, GROBID* Dissertation, in progress

- Building a corpus of dataset and software URLs from **16.1M** scholarly documents, using a hybrid approach combining VisURLX and GROBID for contextualized URL extraction and EnSU for classifying them.

Large-Scale ETD Collection and Metadata Quality | *NER, fuzzy matching, MySQL* [Data]

- Co-built a corpus of **533K+** ETDs from 114 universities via OAI-PMH.
- Built **ETDMIS** to validate and correct metadata with version control, processing **100K** ETDs in 72 hours.
- Designed **ETD-MS v2.0**, a metadata schema with 28 entities and 160 fields.

Los Alamos National Laboratory *Los Alamos, NM*
Research Intern *Jun. 2023 – Aug. 2023*

- Built a transformer-based framework (BERT, DistilBERT, SciBERT) that classifies scholarly URLs in arXiv papers into datasets, software, media, and publishers, with **89% F1**.

Industry Experience

Virginia Marine Resources Commission *Fort Monroe, VA*
Software Engineer Intern *May 2024 – Aug. 2024*

- Redesigned the FOIA website and tracking system, replacing a paper-based process, and tested the 2025 license portal with simulated sales to catch conflicts before launch.

Nascenia *Dhaka, Bangladesh*
Software Engineer *Jan. 2019 – Dec. 2020*

- Built a VAT processing product and a UNDP content management system with Ruby on Rails and MySQL.

Selected Publications

1. **L. Salsabil**, R. R. Obadage, B. Banerjee, S. Alam, W. A. Ingram, E. A. Fox, J. Wu. End-to-End Contextualized URL Extraction from Scholarly Papers. *JCDL 2026.*
2. **L. Salsabil**, R. R. Obadage, B. Banerjee, Y. Abeysinghe, S. Alam, M. Färber, W. A. Ingram, E. A. Fox, J. Wu. Context-Based URL Classification for Open Access Datasets and Software in Scholarly Documents. *JCDL 2025.*
3. R. R. Obadage, **L. Salsabil**, S. Alam, W. A. Ingram, B. Banerjee, E. A. Fox, J. Wu. URL Extraction from Scholarly Documents: A Cross-Format Comparative Analysis. *JCDL 2026.*
4. M. H. Choudhury, **L. Salsabil**, W. A. Ingram, E. A. Fox, J. Wu. ETDPC: A Multimodality Framework for Classifying Pages in Electronic Theses and Dissertations. *AAAI 2024 (IAAI track).*
5. M. H. Choudhury, **L. Salsabil**, H. R. Jayanetti, J. Wu, W. A. Ingram, E. A. Fox. MetaEnhance: Metadata Quality Improvement for Electronic Theses and Dissertations of University Libraries. *JCDL 2023.*
6. **L. Salsabil**, J. Wu, M. H. Choudhury, W. A. Ingram, E. A. Fox, S. M. Rajtmajer, C. L. Giles. A Study of Computational Reproducibility using URLs Linking to Open Access Datasets and Software. *WWW 2022 Companion.*

Honors, Talks, and Service

- **Awards:** NSF Travel Grant, ACM/IEEE JCDL 2026, and Best Short Paper Award, ACM/IEEE JCDL 2023
- **Talks:** JCDL 2026, JCDL 2025, IEEE BigData 2024, ETD 2024, Sci-K Workshop at WWW 2022
- **Reviewer:** JCDL 2026 (4 papers), TPDL 2025, AII-EEKE 2025
- **Graduate Teaching Assistant, ODU:** Data Science (CS 620), HCI (CS 832), Information Literacy (CS 120G)
- **Leadership:** President, ACM-W Chapter, Old Dominion University (2023 – 2024)